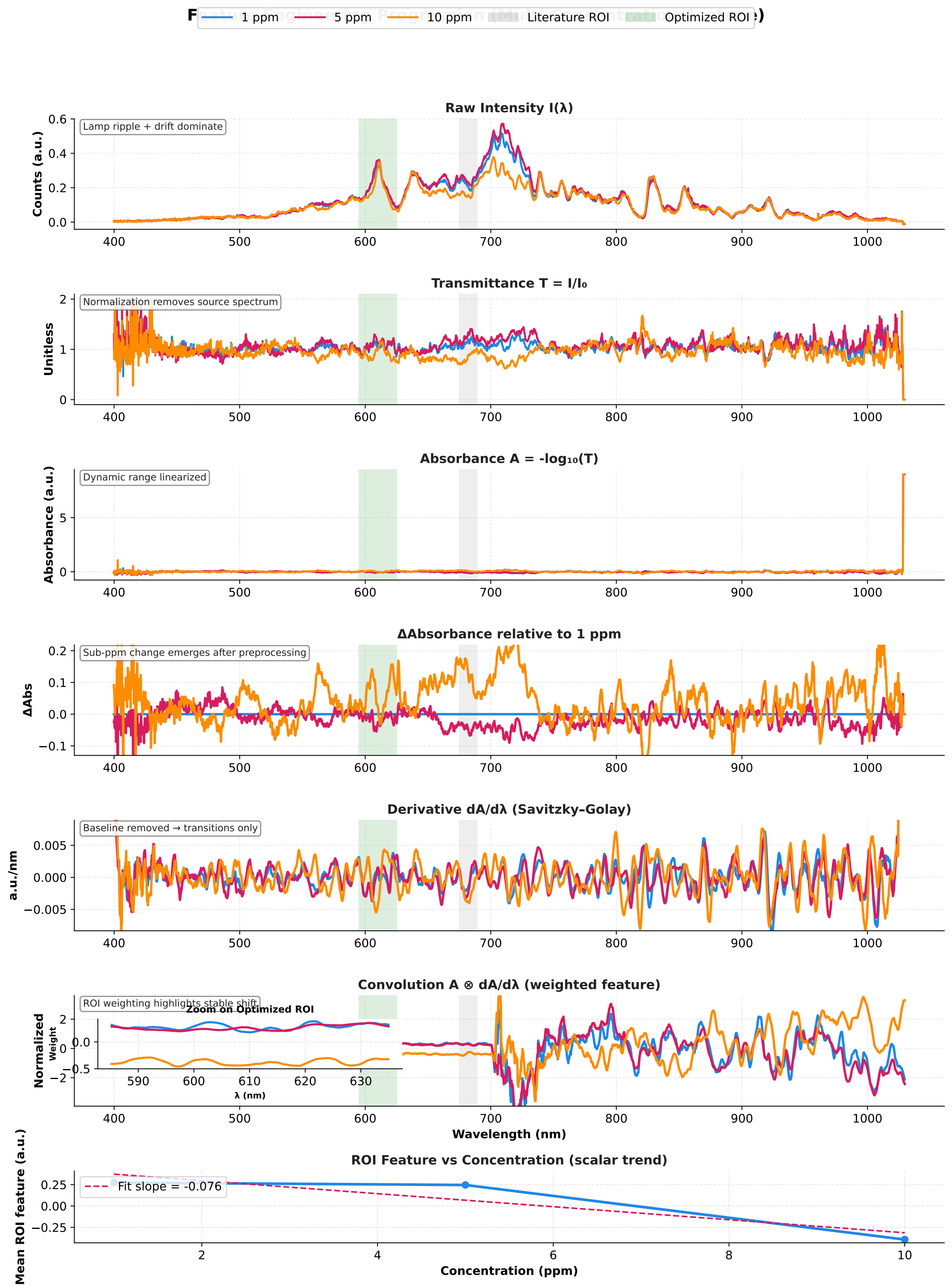
1 ppm
5 ppm
10 ppm
Literature ROI
Optimized ROI
Raw Intensity I(λ)
Lamp ripple + drift dominate
Counts (a.u.)
Transmittance T = I/I₀
Normalization removes source spectrum
Unitless
Absorbance A = -log₁₀(T)
Dynamic range linearized
Absorbance (a.u.)
ΔAbsorbance relative to 1 ppm
Sub-ppm change emerges after preprocessing
ΔAbs
Derivative dA/dλ (Savitzky-Golay)
Baseline removed → transitions only
a.u./nm
Convolution A ⊗ dA/dλ (weighted feature)
ROI weighting highlights stable shift
Zoom on Optimized ROI
Normalized Weight
λ (nm)
Wavelength (nm)
ROI Feature vs Concentration (scalar trend)
Fit slope = -0.076
Mean ROI feature (a.u.)
Concentration (ppm)